

Independent Study

Task 3 AIEA Lab | Apr. 16, 2026

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Objectives

- Deeply exploring LTRAG and setting up their codebase
- Studying my hypothesis and other approaches

What is LTRAG?

LTRAG stands for Logical Reasoning with Thought-Guided RAG. LTRAG is a neurosymbolic method that uses agentic RAG to help solve logic problems. LTRAG compares itself to other methods like LINC, COT, and standard. LTRAG proved to be a more accurate method than the other methods.

LTRAG Framework and Results

On the top of page 3 of the pdf, we can see their framework more clearly. The framework allows for RAG to aid the LLM with translation. Furthermore, RAG on a fixer knowledge base is used to fix errors in translation, which is where LINC struggled. The results are on the top of page 4, showing that LTRAG is the superior method for accuracy.

LTRAG Setup and Changes

I set up the LTRAG codebase on my machine and used Claude to translate their codebase and help me understand it. I still need to finish setting up. LTRAG uses FastGPT for RAG, but I want to switch it to FAISS RAG to make it more repeatable, and also I don't have access to their FastGPT anyway, so I will have to restart things anyway. Hopefully I will have their codebase up and running soon.

Active Research

Mini Hypothesis Experiment: NeSy vs. LRM

This is where this project idea stemmed from. The idea that a normal LLM + NeSy would be better than an LRM. I have started resetting up my codebase from last quarter with the LINC reimplementation. I set up an LLM (4.1-mini) and a similarly capable LRM (5.4-nano) to test my initial mini hypothesis. Next week I will run this mini hypothesis test on actual ProofWriter and FOLIO. Proving this mini hypothesis will be a good first step. Also, this mini experiment will serve as my testing ground for my project.

Other methods

[Logical-SAGE](#) is a paper that improved on LTRAG's RAG and prompts, achieving much better accuracy scores. I could explore their ideas after I am done setting things up.

References

AI use:
I used Claude to translate LTRAG’s comments and strings from Chinese to English. I also used it to explain any preexisting code.

Mini hypothesis: showing LLM and LRM are similarly capable. From [OpenAI's API website](#):

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